

Analysis of Factors Influencing Student Academic Performance in Secondary Schools

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Project Context: Predictive Modeling & Behavioral Analytics Case Study

1. Introduction & Project Objective

The objective of this project is to conduct a comprehensive statistical analysis and develop predictive machine learning models to estimate students' final academic grades (G3) based on demographic, social, and school-related features. The analysis is performed on a dataset containing 649 student profiles from a secondary educational institution.

The core data science challenge lies in identifying key non-academic factors (such as weekly study time, parental education level, or alcohol consumption patterns) that serve as strong determinants of educational success. In compliance with strict predictive modeling practices, the interim period grades (G1 and G2) were completely excluded from the predictor set. This prevents data leakage and ensures that the models rely purely on independent background attributes rather than ongoing short-term performance indicators.

2. Technology Stack & Environment

The entire analytical pipeline was developed in a Python 3.12.9 environment utilizing the following industry-standard libraries:

- **Pandas & NumPy:** For data manipulation, data cleaning, and matrix computations.
- **Matplotlib & Seaborn:** For exploratory data analysis (EDA) and advanced visualizations.
- **Scikit-learn:** For data scaling, train-test splitting, regression modeling, and unsupervised clustering.
- **XGBoost:** For advanced gradient-boosted classification.
- **SciPy & Statsmodels:** For rigorous statistical testing (e.g., Shapiro-Wilk normality testing) and full regression diagnostics.

3. Data Quality Assessment

An initial inspection of the dataset revealed an exceptionally high-quality research sample consisting of 649 records and 33 columns. Key observations include:

- **Missing Values:** The execution of `raw_data.info()` confirmed that the dataset contains no missing values; all 33 columns contain 649 complete, non-null records.

- **Range Verification:** Descriptive statistics indicated that all numerical variables fall within valid, logical ranges (e.g., student age ranges from 15 to 22 years, and historical attributes fit expected scales).
- **Target Variable:** The final grade (G3) takes discrete values in the range of 1 to 10. The sample mean sits at approximately 6.22, with a standard deviation of 1.59.

4. Data Cleaning & Transformation

The data preparation pipeline was split into two crucial phases:

- **Phase 1: Feature Selection:** In accordance with the project constraints, the G1 and G2 columns were entirely dropped. Retaining them would introduce severe data leakage, artificially inflating model accuracy based on immediate past performance rather than structural background attributes.
- **Phase 2: Binary Variable Encoding:** To allow mathematical algorithms to process categorical text data, binary variables (features with exactly two categories, such as "yes/no" or "Urban/Rural") were mapped into a 0/1 numeric format.

Manual mapping was explicitly chosen over automated encoders to maintain absolute control over coefficient interpretation (e.g., mapping a value of 1 directly to the explicit desire to pursue higher education – higher).

5. Nominal Variable Encoding

Features containing more than two distinct categories (such as parental occupations: Mjob, Fjob, or the reason for choosing the school: reason) were transformed using One-Hot Encoding. This method generates separate binary columns for each unique category, preventing the models from incorrectly assuming an artificial numerical hierarchy or ordering (which would inadvertently occur if labeled simply as 1, 2, or 3). Following this operation, the feature space in the training set expanded to 44 predictors.

```
# Mapping binary variables to numerical format (0/1)
# Manual mapping ensures full control over the interpretation of variable direction
binary_features = {
    'schoolsup': {'yes': 1, 'no': 0}, 'famsup': {'yes': 1, 'no': 0},
    'paid': {'yes': 1, 'no': 0}, 'activities': {'yes': 1, 'no': 0},
    'nursery': {'yes': 1, 'no': 0}, 'higher': {'yes': 1, 'no': 0},
    'internet': {'yes': 1, 'no': 0}, 'romantic': {'yes': 1, 'no': 0},
    'school': {'GP': 1, 'MS': 0}, 'sex': {'F': 1, 'M': 0},
    'address': {'U': 1, 'R': 0}, 'famsize': {'GT3': 1, 'LE3': 0},
    'Pstatus': {'T': 1, 'A': 0}
}

for col, mapping in binary_features.items():
    df_processed[col] = df_processed[col].map(mapping)
```

6. Train-Test Split & Feature Synchronization

To ensure an unbiased evaluation of the models' generalization capabilities, the dataset was partitioned into two isolated subsets:

- **Training Set (80%):** 519 student profiles.
- **Test Set (20%):** 130 student profiles.
- **Seed for Reproducibility:** A seed with a value of "1234" was used

A strict column synchronization step was implemented after the split. This data engineering best practice ensures that the test set inherits the exact same feature schema as the training set, preventing runtime failures if a rare category present in the training data does not appear in the smaller test sample.

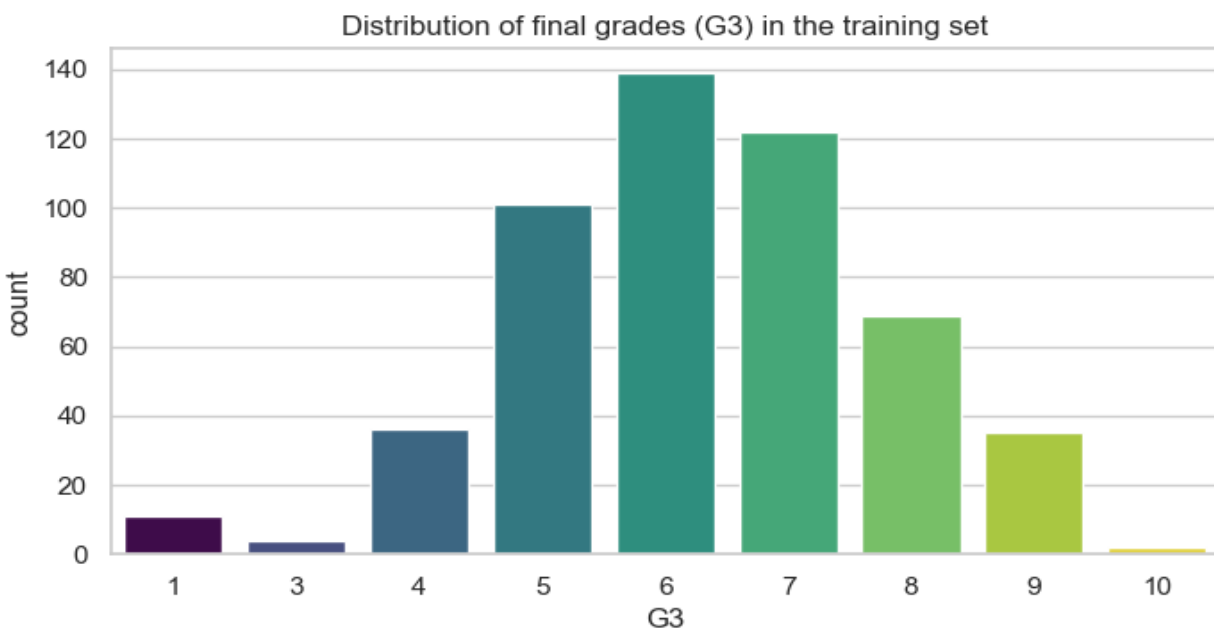
```
# Splitting into training and test sets before categorical transformations
# This avoids information leakage about rare categories from the test set
train_raw, test_raw = train_test_split(df_processed, test_size=0.2, random_state=SEED)
```

7. Exploratory Data Analysis (EDA)

Exploratory analysis was conducted exclusively on the training subset (519 observations). This boundary prevents data leakage, ensuring that model architecture decisions are not subconsciously influenced by structural trends present in the testing subset.

7.1. Distribution of the Target Variable (G3)

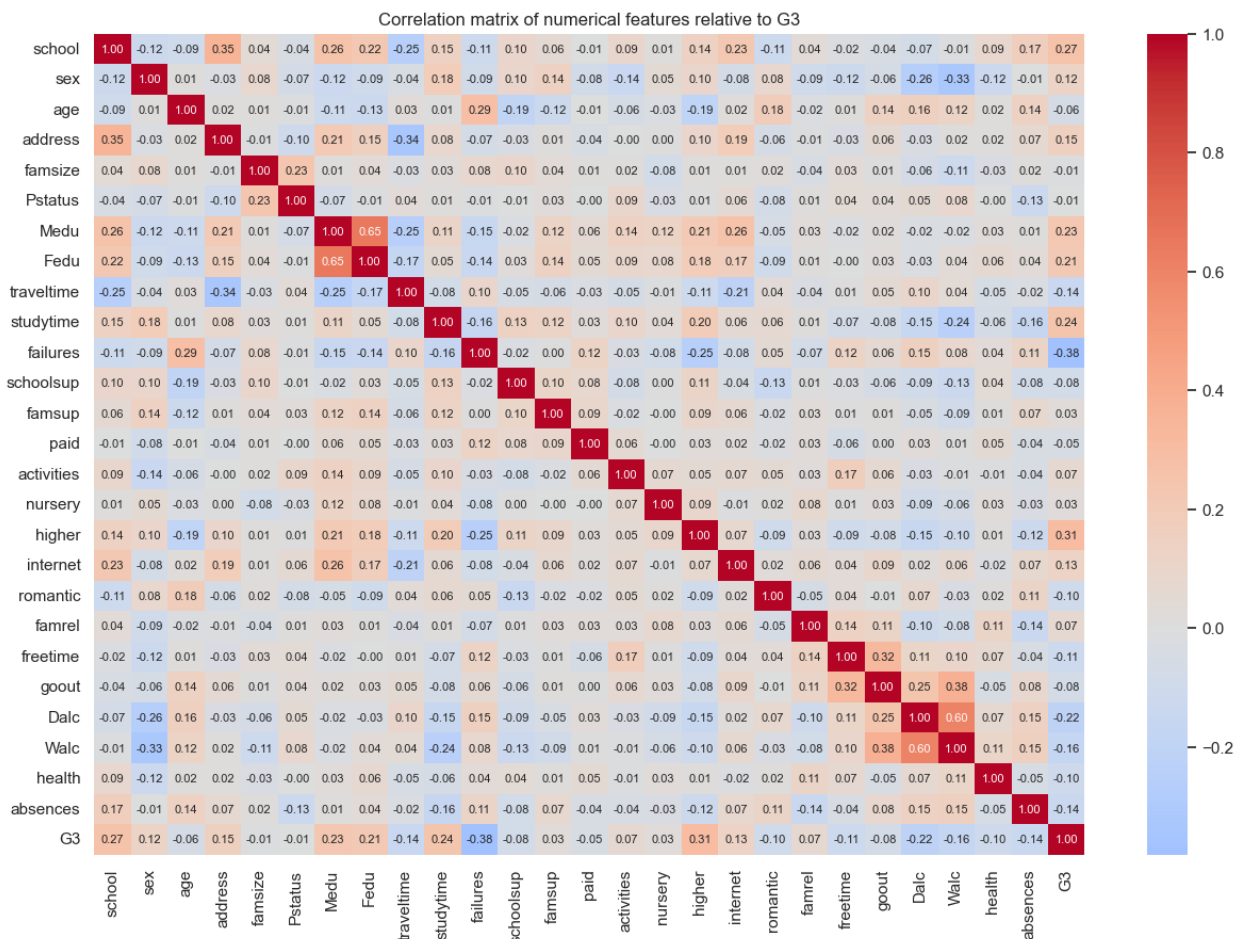
An initial histogram analysis of G3 revealed a primary mode at grade 6 (~140 students), with mid-scale grades (5, 6, 7) making up the vast majority of the dataset. While the distribution roughly follows a bell-shaped curve, a distinct statistical anomaly was identified at grade 1 (10 observations), contrasting with a near-zero frequency for grades 2 and 3. This suggests a subset of students who either entirely withdrew from the final evaluation or experienced catastrophic academic failure.



7.2. Linear Correlation Analysis

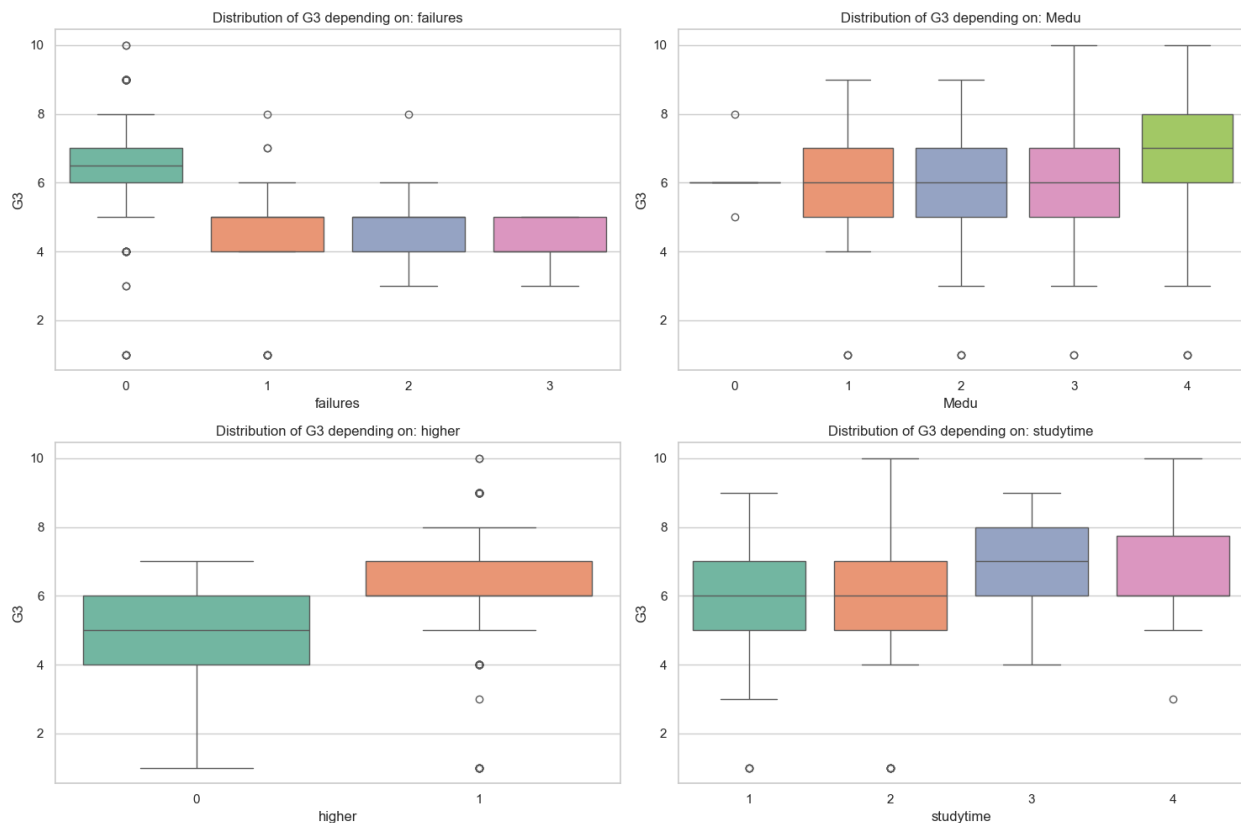
A Pearson linear correlation matrix was calculated to identify single-feature linear relationships with G3:

- **Strongest Linear Inhibitor:** The variable failures (number of past class failures) displays the strongest negative correlation with the final grade (approximately -0.38). As past academic setbacks increase, the final grade drops significantly, making it a critical predictor.
- **Key Supporting Factors:** Significant positive correlations were observed across several primary predictors. The strongest positive correlation with the final grade is exhibited by higher (0.31), reflecting that a student's internal motivation to pursue higher education is the leading linear driver of success. This is followed by school (0.26), indicating a measurable institutional baseline effect. Additionally, structural personal and environmental factors showed solid positive correlations, namely studytime (0.24), representing weekly hours dedicated to studying, and Medu (0.23), representing the mother's education level. These findings validate that personal work ethic, institutional context, and household cultural capital exert a distinct, cumulative linear impact on performance.
- **Weak Linear Features:** Most remaining demographic and lifestyle variables showed standalone correlations below +/-0.20. While minor individually in a linear context, their complex interactions are better leveraged in non-linear models.



7.3. Feature Insight via Boxplot Analysis

- **Past Failures (failures):** A massive performance gap exists between students with zero past failures and those with one or more. For failures = 0, the median grade hovers around 6–7, whereas any value above zero triggers a sharp drop in the median to 4–5.
- **Educational Aspirations (higher):** Students aiming to pursue higher education exhibit significantly higher and more stable grades. A lack of higher education aspirations correlates with a lower median and much wider variance.
- **Study Time (studytime):** A clear upward trend appears across categories 1, 2, and 3. Interestingly, category 4 (maximum study time) does not yield a higher average than category 3, but it substantially compresses the distribution, drastically minimizing the occurrence of failing grades.



8. Estimation Model (G3 as a Continuous Variable)

To treat the final grade as a continuous metric, a classical Ordinary Least Squares (OLS) Linear Regression model was trained with an intercept (fit_intercept=True) on scaled data.

8.1. Feature Scaling

Because input features utilize entirely different units (e.g., absences ranging from 0 to 32, and studytime on a scale of 1 to 4), a StandardScaler was applied. This centered all variables to a mean of 0 and a standard deviation of 1, rendering the regression weights directly comparable and interpretable.

8.2. Continuous Performance Evaluation

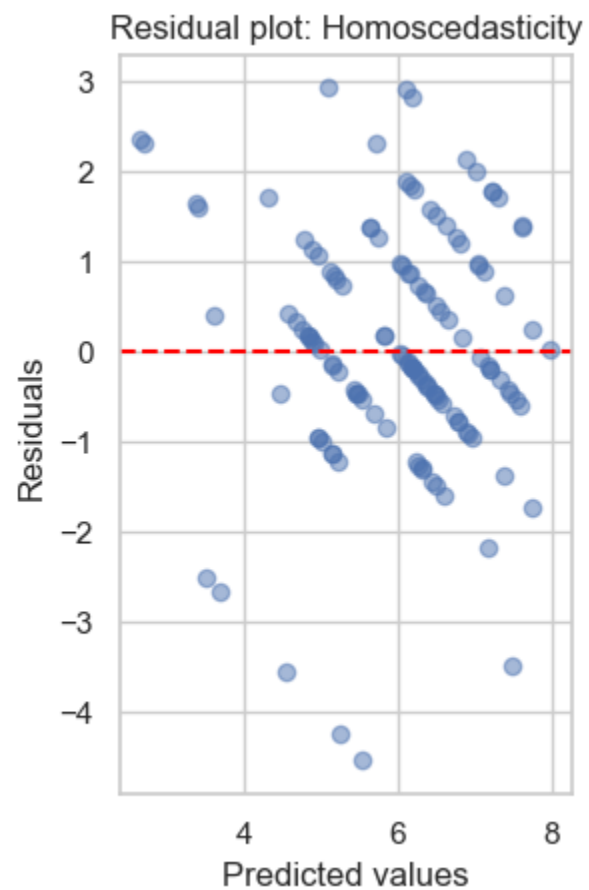
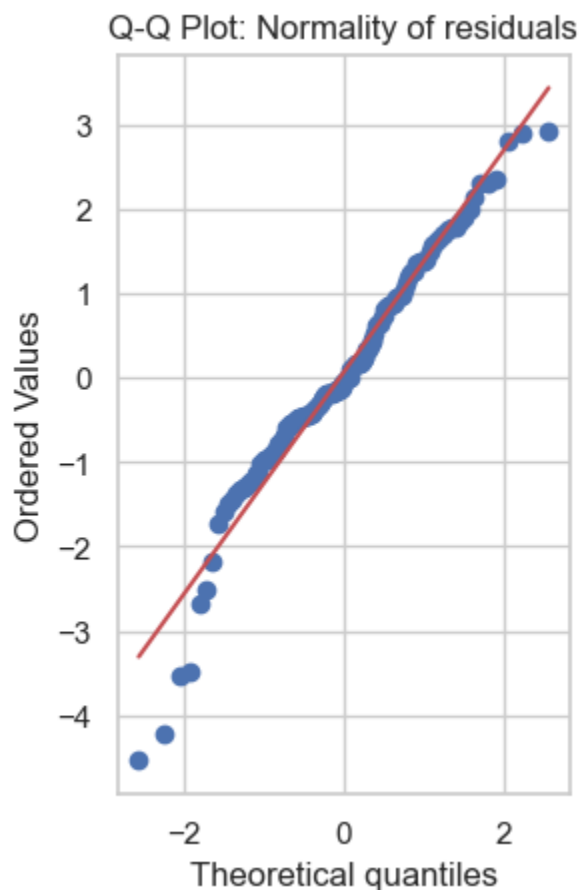
Continuous model outputs were rounded to the nearest integer to calculate exact classification-aligned metrics on the test set:

```
LINEAR REGRESSION MODEL EVALUATION (Estimation)
--- Results for model: Linear Regression ---
Accuracy:                0.3769
Accuracy (Margin +/- 1): 0.7846
Mean Absolute Error MAE: 0.9987
```

Interpretation: While exact accuracy sits at 26.15%, nearly three-quarters of the model's predictions fall within a single grade point of the true value. An MAE of 1.13 on a 1–10 scale demonstrates that the linear model successfully captures the general academic standing of a student.

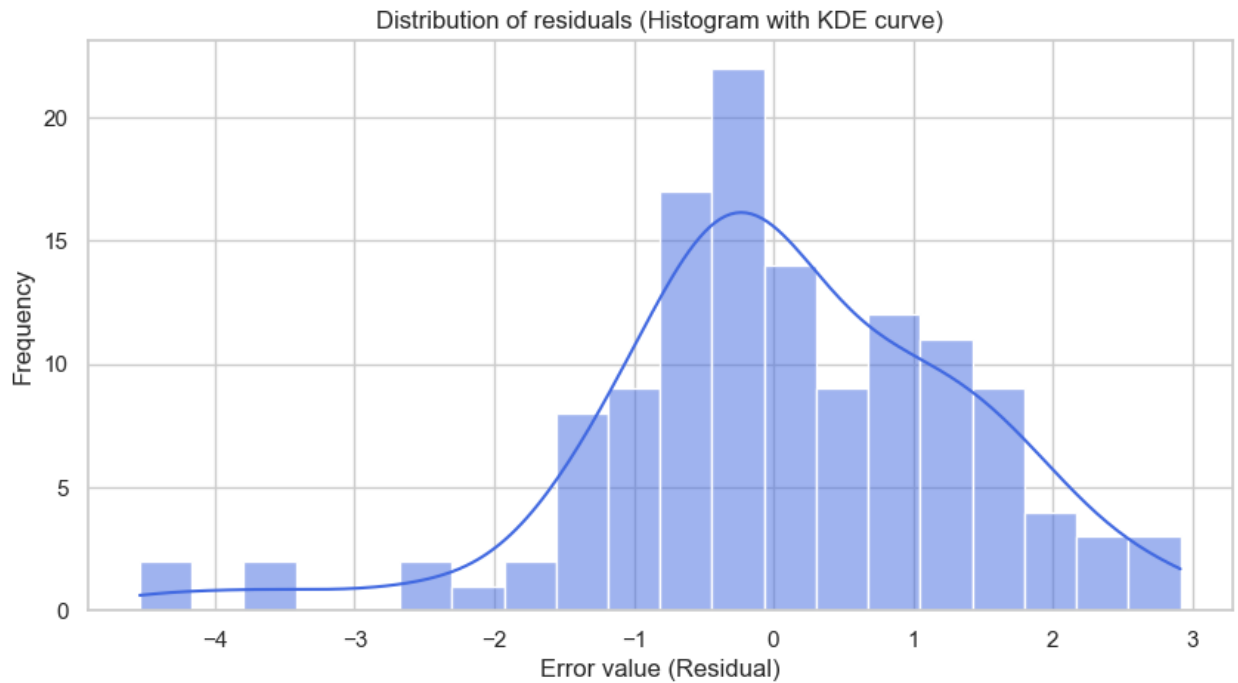
8.3. Regression Diagnostics & Assumption Verification

- **Residual Normality:** A Q-Q plot of the regression residuals showed noticeable deviations from the theoretical line, particularly in the lower quartile. A formal Shapiro-Wilk test returned a p-value of 0.0016. Since $p < 0.05$, the hypothesis of perfect error normality is rejected. This behavior is structural, caused by the discrete integer nature of the true target variable.

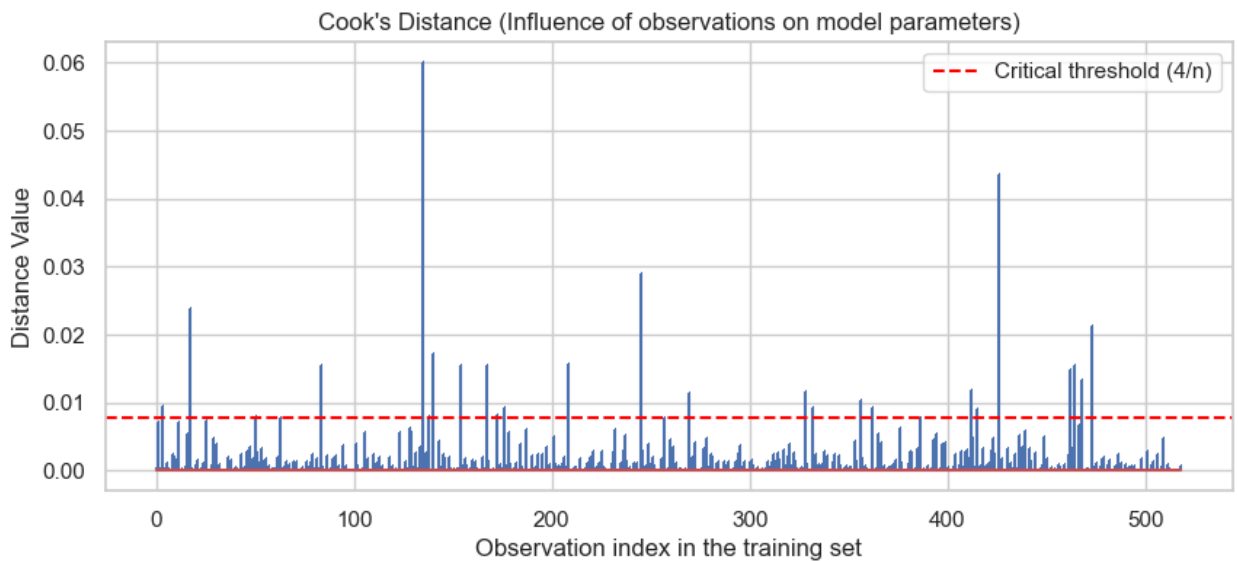


- **Error Distribution & Heteroscedasticity:** A residual histogram confirmed a severe left-skew. While the majority of errors cluster tightly around zero, a subset of large negative residuals

represents instances where the model heavily overpredicted the student's grade. Crucially, the Durbin-Watson statistic recorded an excellent value of 2.0353, confirming that the errors are independent and entirely free from autocorrelation issues.



- **Influential Observations (Cook's Distance):** Cook's Distance identified 24 observations exceeding the critical operational threshold of $4/n$. These extreme outliers exert disproportionate leverage on the regression line. They represent "atypical" student profiles whose background strongly indicates high success, yet their actual final grade was a 1, placing a mathematical floor on the minimum achievable MAE.



8.4. Analysis of Regression Weights

The standardized coefficients revealed the direct linear impact of features while controlling for all other variables, allowing for the identification of the most statistically significant predictors within the system.

- **Top Positive Drivers:** The model identifies several key features that increase the predicted final G3 grade. Institutional affiliation (specifically to the 'GP' facility) emerges as the most significant positive structural driver of the final grade with a weight of +0.2907. This is followed by the explicit desire to pursue higher education (+0.2357), which remains a robust positive stimulant, reinforcing the role of student ambition. Other influential positive factors include gender (+0.1573), father's education level (+0.1568), and the number of hours spent studying (+0.1525), all of which translate directly and linearly into higher marks.
- **Top Negative Inhibitors:** Certain features exert a significant downward pressure on the predicted final grade. The number of past class failures (-0.4134) is by far the heaviest anchor on academic performance; each historical failure drastically forces down the continuous G3 prediction. The model also identifies that extra school-provided academic support (-0.2394) is strongly correlated with students who are already struggling and require intervention, rather than being a cause of low grades itself. Furthermore, higher truancy rates represented by absences (-0.1937), self-reported current health status (-0.1678), and weekday alcohol consumption (-0.1065) act as additional negative inhibitors to the final grade.
- **Diagnostic Multicollinearity Notes:** During model evaluation, a warning was triggered regarding highly correlated features, specifically identifying the guardian_mother variable. This signals a presence of localized multicollinearity within the family-related predictor subset. However, from a domain-interpretability standpoint, this variable was retained to maintain full baseline transparency. Stability testing confirmed that its presence exerts a negligible impact on the variance and mathematical stability of the core structural weights, such as failures and higher, allowing for reliable interpretation of the primary coefficient rankings.

9. Classification Model Characteristics

In the second approach to the research problem, the final grade G3 was treated as a qualitative variable (a class). Instead of predicting a value on a numerical axis, the model attempts to assign each student to one of the categories (grades 1 to 10). The selected algorithm is XGBoost (Extreme Gradient Boosting).

Why XGBoost?

- **Non-linearity:** Unlike linear regression, XGBoost can detect complex, non-linear dependencies between features (e.g., a situation where a high number of absences is detrimental only when the student lacks support at home).
- **Robustness:** This algorithm handles various data types (binary, numerical, categorical) exceptionally well and is resistant to outliers identified during the EDA phase (e.g., the unusual frequency of grade 1).
- **Ensemble Learning:** It operates by building multiple weak models (decision trees) that sequentially correct the errors of their predecessors.

The XGBoost model was trained with the following parameters: `n_estimators=100`, `learning_rate=0.3`, `max_depth=6`, and the multi:softmax objective function (`eval_metric='logloss'`).

9.1. Classifier Quality Metrics

The model was evaluated on the test set using the same metrics as the regression model to allow for a direct comparison.

- **Accuracy (Exact):** 32.31%
- **Accuracy (Margin +/- 1):** 78.46%
- **Mean Absolute Error (MAE):** 1.0538

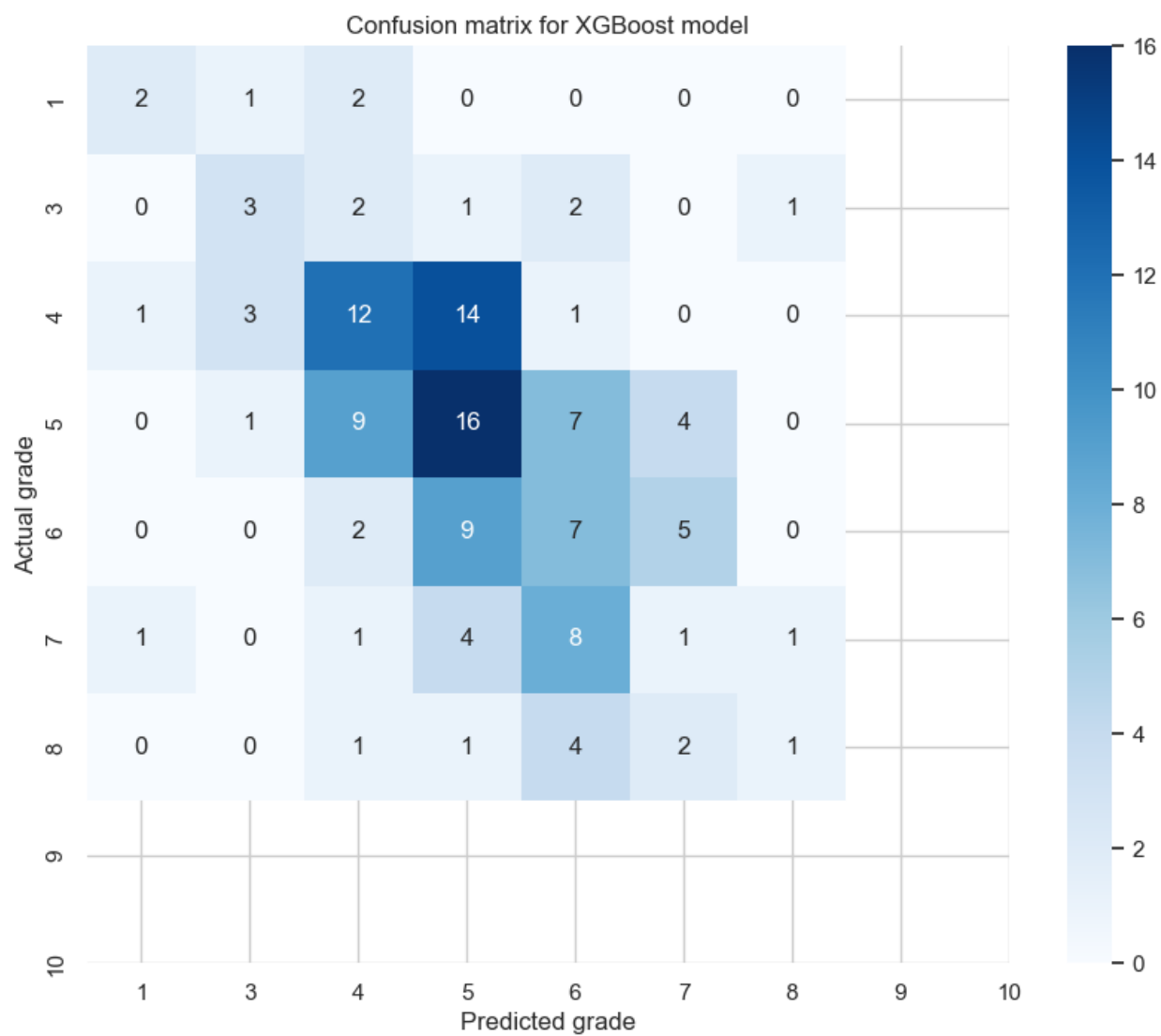
```
XGBOOST MODEL EVALUATION (Classification)
--- Results for model: XGBoost Classification ---
Accuracy:                0.3231
Accuracy (Margin +/- 1): 0.7846
Mean Absolute Error MAE: 1.0538
```

Conclusions from the results: The classification model achieved better precision than the linear regression model. The MAE value dropped to 1.05, which suggests that the non-linear approach allowed for a more accurate "hit" on specific grades. Crucially, the accuracy within a +/- 1 error margin remained high at 78.46%, confirming the stability of the predictions.

9.2. Error Analysis (Confusion Matrix)

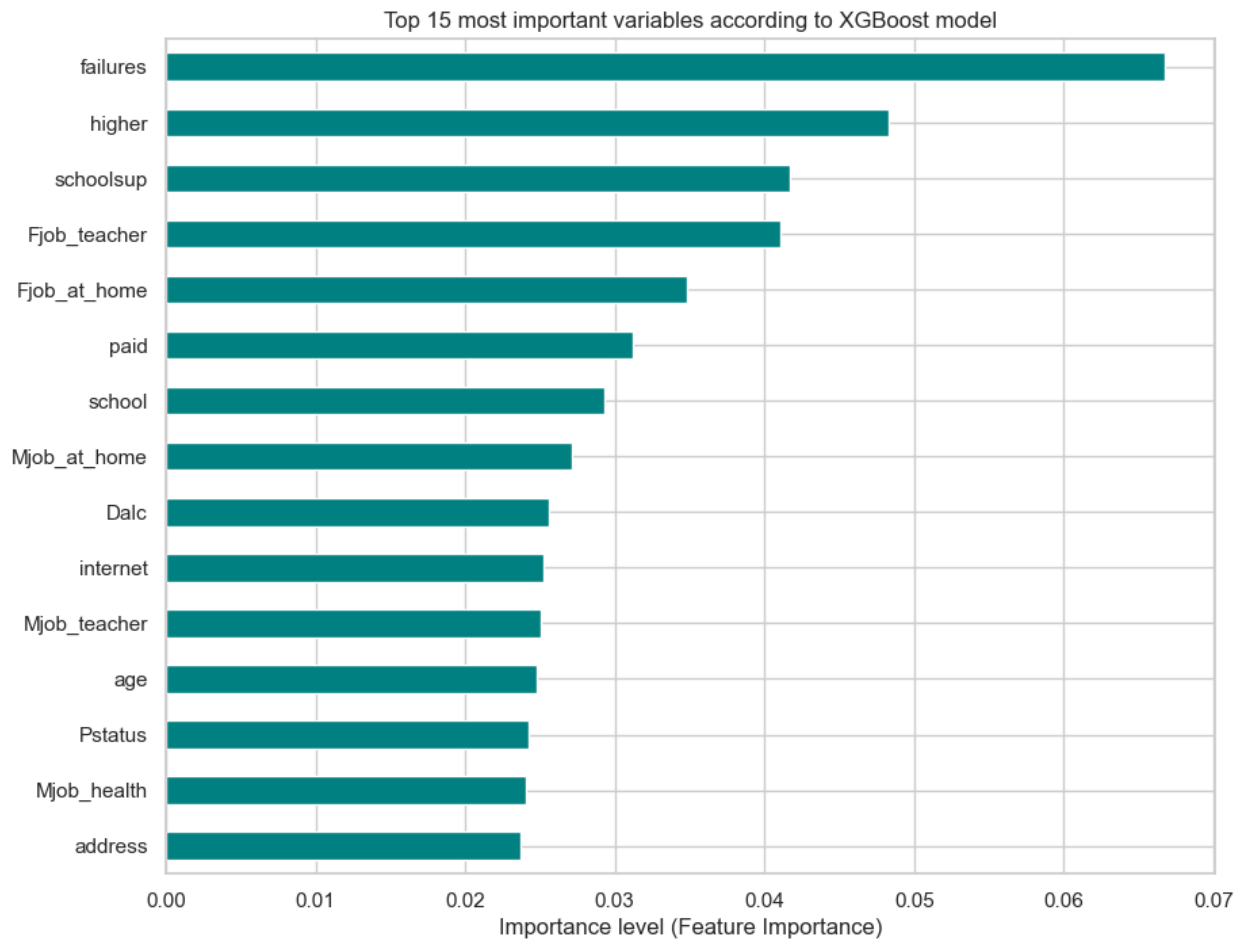
Visualizing the model's accuracy in the form of a confusion matrix allows for a precise assessment of classification quality for individual grades.

- **Result Concentration:** The highest values cluster along the main diagonal of the plot, which indicates high model accuracy. Errors are localized in nature – the model most often misses by only one grade point (e.g., classifying a grade of 11 as 10 or 12).
- **Difficult Areas:** The matrix shows greater error dispersion in the middle range of the scale. This is due to the fact that this group is the most heavily represented in the dataset, which naturally increases the number of model errors in this interval. At the same time, the model shows difficulties in precisely classifying the lowest grades (0–5), where errors are more dispersed due to the lower predictability of results for at-risk students.



9.3. What Determines Student Success According to the Model?

One of the greatest advantages of XGBoost is the ability to generate a Feature Importance plot, which shows which variables most frequently determined the classification of students into groups with different grades.



9.4. Comparison with EDA and Regression Analysis

The feature importance analysis in the XGBoost model largely confirms the conclusions drawn during the data exploration stage but sheds new light on certain variables:

- **Most Important Predictor (failures):** In accordance with the assumptions from the EDA stage (correlation -0.41) and the weights from the regression model, the failures variable remains the undisputed most important predictor. Previous school failures are the strongest warning signal.
- **Determination (higher):** The desire to continue education at university (higher) holds second place. This confirms that a student's internal motivation is crucial for the final G3 result.
- **New Significant Features:** XGBoost highlighted variables that were less visible in linear regression:
 - **schoolsup and Fjob_teacher:** Extra school-provided academic support and the father's profession as a teacher proved to have high discriminatory power for the model.

9.5. Classification Stage Summary

XGBoost proved that it can better parameterize the problem of student grading than a linear model. Although the difference in accuracy is not massive, the lower MAE indicates fewer glaring errors. The model effectively combined hard data on failures with subtler signals, such as parental profession or the student's academic support status.

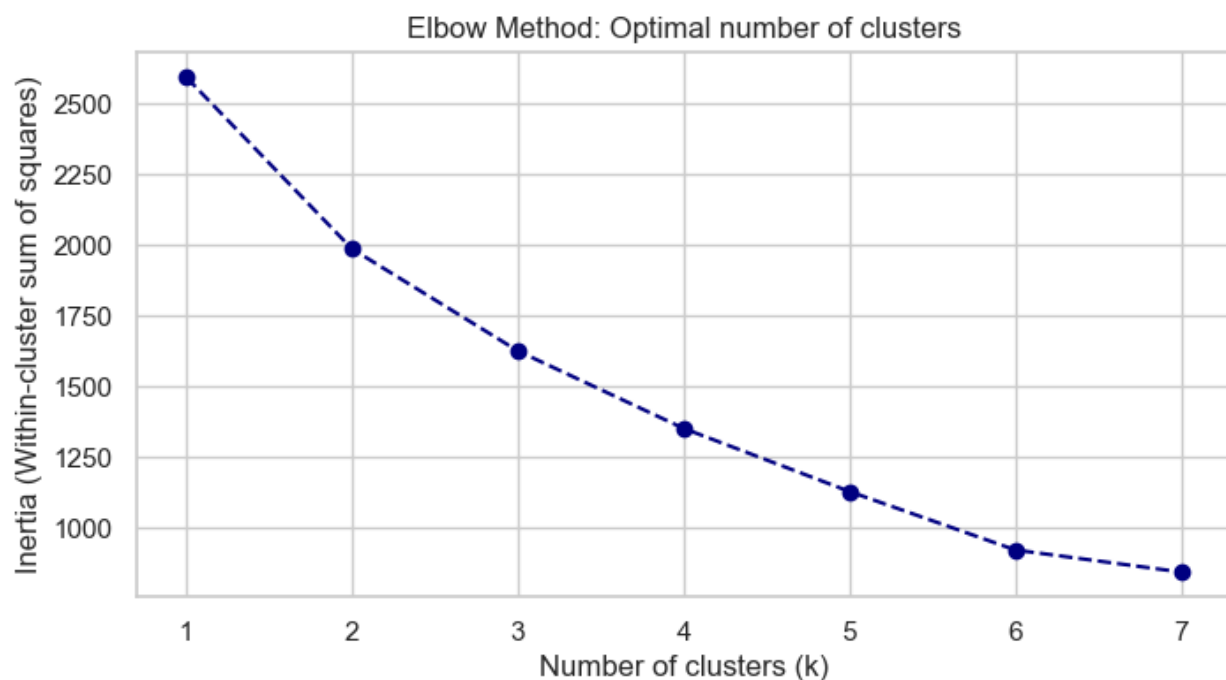
10. Unsupervised Clustering (K-Means)

The final stage of the analysis involved conducting unsupervised machine learning using the K-Means algorithm. The objective was not to predict a specific grade, but to segment students into groups sharing similar socio-educational characteristics.

- **Feature Selection:** Five variables, identified as most significant in earlier stages (EDA, XGBoost), were selected for the model: failures, absences (indicators of difficulty and engagement), Medu, studytime, and higher (indicators of cultural capital and ambition). Utilizing only the most critical features prevents information "noise" generated by variables with low predictive power.
- **Algorithm Configuration:** The K-Means algorithm was applied with three clusters ($k=3$), k-means++ initialization, and a fixed random seed.

10.1. Optimization of the Number of Groups (Elbow Method)

To determine the optimal number of clusters (k), the Elbow Method was employed by analyzing the reduction in inertia (the sum of squared distances within clusters).



Graph Interpretation: The inertia plot shows a smooth decline without a sharply defined "elbow point." Given the lack of a distinct inflection in the curve, $k=3$ was selected. This cluster count provides optimal reporting clarity and enables an intuitive division of students into three fundamental outcome groups, despite the absence of rigid mathematical dictates for this specific value.

10.2. Characteristics of Identified Student Profiles

Based on the mean feature values within the training set, three distinct student archetypes were identified:

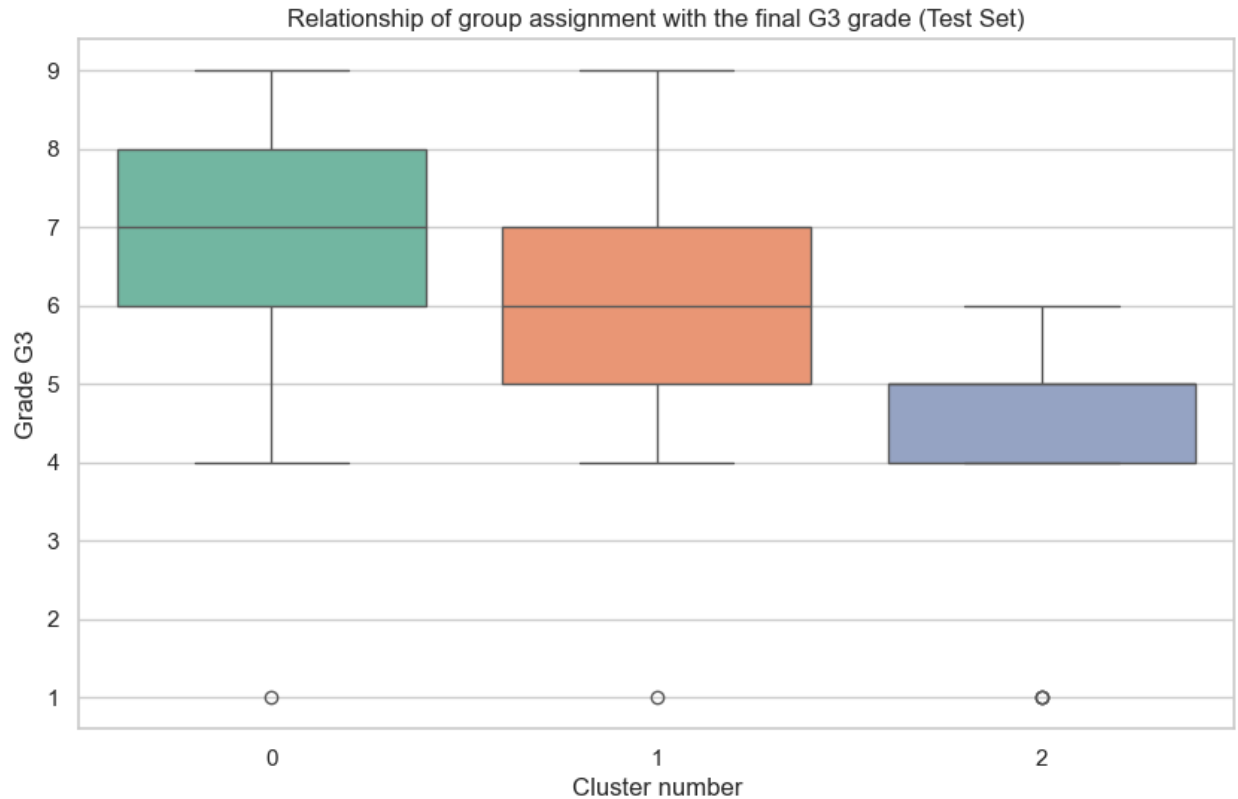
Feature	Group 0: "Ambitious Leaders"	Group 1: "Stable Average"	Group 2: "At-Risk Students"
Mean G3	6.68	6.17	4.78
Failures	0.08	0.15	0.84
Absences	3.73	3.27	5.33
Higher	100% (Yes)	100% (Yes)	~10% (No)
Medu	3.57	1.57	1.90

Profile Descriptions:

- **Group 0 ("Ambitious Leaders"):** Students with the best domestic background (high maternal education) and the lowest failure rate. All plan to pursue higher education, resulting in the highest academic performance.
- **Group 1 ("Stable Average"):** Students with lower domestic capital but high determination (all intend to study). Despite fewer resources, they maintain a solid average due to a low number of absences.
- **Group 2 ("At-Risk Students"):** The highest-risk group, characterized by a lack of academic ambition (higher $\approx 10\%$), a high number of missed classes, and frequent past failures. Their average grade is drastically lower than that of other groups.

10.3. Do Clusters Actually Differentiate Grades?

To verify the utility of the clustering model, the G3 grade distribution was analyzed for clusters assigned to the test set data (which the model had not seen during training).



- **Divisional Effectiveness:** The results confirm the logic of the clustering. The mean G3 grade for Group 0 in the test set is 6.70, while Group 2 averages only 4.42.
- **Stability:** Group 1 ("Stable") maintains a repeatable outcome (6.08), proving that the model successfully separated "average" students from both top performers and those experiencing difficulties.
- **Test Set Distribution:** Analysis of the test set reveals distinct grade ranges:
 - **Group 0:** Grades range from 4 to 9 (1st quartile: 6, median: 7, 3rd quartile: 8).
 - **Group 1:** Grades range from 4 to 9 (1st quartile: 5, median: 6, 3rd quartile: 7).
 - **Group 2:** Grades range from 4 to 6 (1st quartile: 4, median: 4, 3rd quartile: 5).
- **Outliers:** Boxplots indicate that low grades (even 0/1) occur in every group. This means that while a cluster profile provides a strong indication of a likely outcome, individual random factors still play a significant role.

11. Project Summary

The project encompassed a complete data analysis lifecycle – ranging from cleaning and exploratory analysis to advanced predictive modeling and unsupervised segmentation.

Key Achievements:

- **Identification of Primary Success Determinants:** The number of past academic failures (failures) and educational ambitions (higher) were confirmed as the primary drivers of student performance.
- **Model Comparison:** Surprisingly, the Linear Regression (Estimation) model demonstrated slightly better performance in this specific iteration (MAE=0.9987) compared to the XGBoost Classifier (MAE=1.0539). The models exhibited identical stability within a +/-1 degree margin (78.46%).
- **Student Profiling:** The K-Means algorithm successfully automated the detection of student groups at risk of low performance, which, in a practical educational environment, could facilitate early pedagogical intervention.